SQL ASSIGNMENT – 1

Scenario 1: Student Management System

A college wants to maintain student details in a database.

Table Name: Students

Columns:

- Student_ID
- Student_Name
- Age
- Department
- City
- Marks

Questions:

1. Create a database named CollegeDB.
2. Create a table named Students with the above columns.
3. Insert the following 5 student records into the table.

Student_ID	Student_Name	Age	Department	City	Marks
101	Arun	21	Python	Chennai	85
102	Priya	22	Java	Madurai	90
103	Kumar	20	Python	Coimbatore	75
104	Divya	21	SQL	Chennai	95
105	Rahul	23	Java	Salem	80

4. Display all columns from the Students table.
5. Display only Student_Name and Department.
6. Display only Student_Name and Marks.
7. Display all student records in ascending order of Marks.
8. Display all student records in descending order of Marks.
9. Display student names in alphabetical order.
10. Display student records in descending order of Student_ID.

Answers:

1. Create a database named CollegeDB.

   Query:

   CREATE DATABASE College_db;

2. Create a table named Students with the above columns.

   Query:
   ```
   CREATE TABLE Students(
   Student_ID int,
   Student_Name varchar(30),
   Age int,
   Department varchar(30),
   City varchar(15),
   Marks int);
   ```

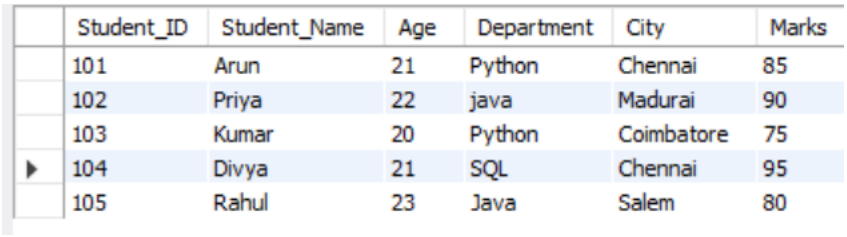
3. Insert the following 5 student records into the table.

   Query:
   ```
   INSERT INTO students values
   (101,'Arun',21,'Python','Chennai',85),
   (102,'Priya',22,'java','Madurai',90),
   (103,'Kumar',20,'Python','Coimbatore',75),
   (104,'Divya',21,'SQL','Chennai',95),
   (105,'Rahul',23,'Java','Salem',80);
   ```

4. Display all columns from the Students table.

   Query:
   SELECT * FROM Students;

   Output:

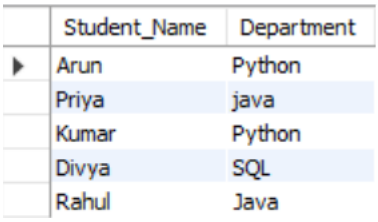


Student_ID	Student_Name	Age	Department	City	Marks
101	Arun	21	Python	Chennai	85
102	Priya	22	java	Madurai	90
103	Kumar	20	Python	Coimbatore	75
104	Divya	21	SQL	Chennai	95
105	Rahul	23	Java	Salem	80

5. Display only Student_Name and Department.

   Query:
   SELECT Student_Name,Department FROM Students;

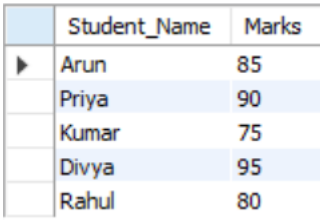
   Output:



Student_Name	Department
Arun	Python
Priya	java
Kumar	Python
Divya	SQL
Rahul	Java

6. Display only Student_Name and Marks.
   Query:
   SELECT Student_Name,Marks FROM Students;

   Output:



Student_Name	Marks
Arun	85
Priya	90
Kumar	75
Divya	95
Rahul	80

7. Display all student records in ascending order of Marks
   Query:
   SELECT * FROM Students
   ORDER BY Marks ASC;

Output:

Student_ID	Student_Name	Age	Department	City	Marks
103	Kumar	20	Python	Coimbatore	75
105	Rahul	23	Java	Salem	80
101	Arun	21	Python	Chennai	85
102	Priya	22	java	Madurai	90
104	Divya	21	SQL	Chennai	95

8. Display all student records in descending order of Marks.

Query:

```
SELECT * FROM Students
ORDER BY Marks DESC;
```

Output:

Student_ID	Student_Name	Age	Department	City	Marks
104	Divya	21	SQL	Chennai	95
102	Priya	22	java	Madurai	90
101	Arun	21	Python	Chennai	85
105	Rahul	23	Java	Salem	80
103	Kumar	20	Python	Coimbatore	75

9. Display student names in alphabetical order.

Query:

```
SELECT Student_Name FROM Students
ORDER BY Student_Name ASC;
```

Output:

Student_Name
Arun
Divya
Kumar
Priya
Rahul

10. Display student records in descending order of Student_ID.

Query:

```
SELECT * FROM Students
ORDER BY Student_ID DESC;
```

Output:

Student_ID	Student_Name	Age	Department	City	Marks
101	Arun	21	Python	Chennai	85
102	Priya	22	java	Madurai	90
103	Kumar	20	Python	Coimbatore	75
104	Divya	21	SQL	Chennai	95
105	Rahul	23	Java	Salem	80